Syracuse University
School of Information Studies

Week 10
Association Rules Mining

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Announcements

Exam Date

- **7th April 2026: Undergrad, and**
- **9th April: Grad**
- The exam will be administered in lab sections, on paper.
- Covers **Weeks 1-9**
- No computer, just a pen/pencil and a hard copy of your notes (e.g.: printed/hand-written notes or copies of the textbook, printed HW submissions, lecture slides, or printed important concepts/commands.)
- There will be different versions of the exam.
- You will have ~75 minutes to complete the exam.
- The exam will contain 8 multiple choice questions, and 12 conceptual R coding questions.

Announcements (only about Lecture)

- **2nd April** (Next Thursday): **No Lecture, Study for Exam**
- **9th April (Next Next Thursday): No Lecture, Video Tutorial will be posted to help with Lab 11 and HW 11**
- **16th April (We meet): Last Lecture of the course (Week 12)!!**

Keep an eye for the Lab and HW deadlines!

An Example of Unsupervised Learning: Association Rules Mining

- Association rules are **"if–then" statements** that show relationships between items that often appear together.
- For example:
  - **If** a person buys pasta,
  - **Then** they are also likely to buy pasta sauce.

- **Why it's useful:**
  It helps find **patterns or connections** in large sets of data.

Sparse Matrix Representation

	Wipes	Beer	Diapers	Lettuce	Flakes	Milk
Cart 1		1	1	1		1
Cart 2	1			1		1
Cart 3		1	1		1	1

Item set Representation

Cart1: 2,3,4,6
Cart2: 1,4,6
Cart3: 2,3,5,6

Association Rules Mining

Support:

- Tells **how common** a combination of items is in all transactions.
- **Support(A ∪ B) = (Number of transactions that contain both A and B) ÷ (Total number of transactions).**
- **Example:**
  If 2 out of 10 customers buy both **milk and bread**,
  then support = 2/10 = **20%**.

Confidence:

- Tells **how likely** someone is to buy item B **when** they buy item A.
- Formula:
  Confidence(A → B) = (Transactions with both A and B) ÷ (Transactions with A)

- **Example:**
  If 8 people buy milk, and 6 of them also buy bread, confidence = 6/8 = **75%**.

Lift: How Much Does LHS (If) Boost the Purchase of RHS (Predict)?

What the Value Means:

- **Lift > 1:** The presence of LHS **increases** the probability of buying RHS.
- **Lift = 1:** The items are independent. Buying LHS has **no effect** on buying RHS.
- **Lift < 1:** The presence of LHS **decreases** the probability of buying RHS (a negative correlation).

Conviction: How Incorrect are the Rule's Predictions?

- **Conviction** measures how often the rule makes a **wrong prediction**.
- A perfect rule (**Confidence=1**) never predicts the wrong thing.
- **High Conviction** indicates the RHS is highly dependent on the LHS.
- **Low Conviction** means the items are independent.

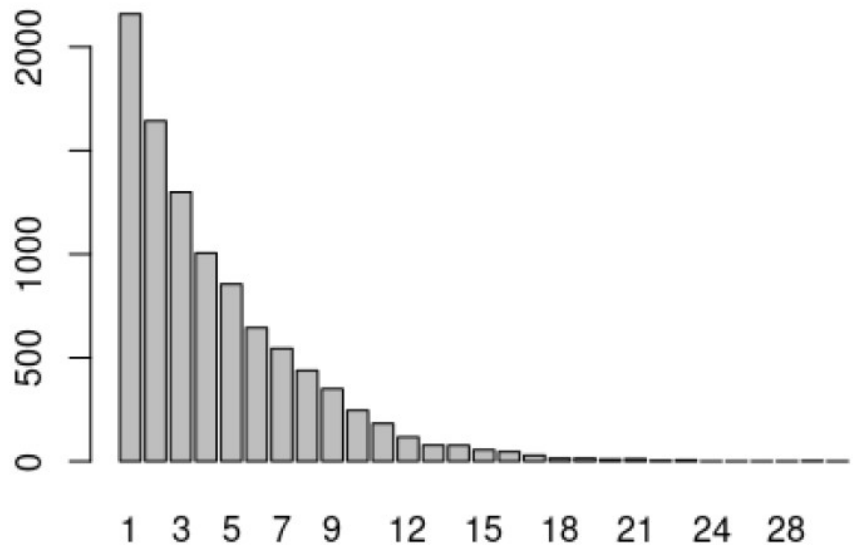
Coding

```
#install.packages("arules")

library("arules")
```

```
data (Groceries)
myGroc <- Groceries # Make a copy for safety
str(myGroc)
```

```
barplot(summary(myGroc)@lengths)
```



- **Groceries** is a dataset of every single customer receipt (your sales data).

- **barplot(summary(myGroc)@lengths):**
  - **summary(myGroc)** goes through every receipt and counts how many items are in it (e.g., Receipt 1 has 3 items, Receipt 2 has 5 items, etc.). The result of that summary is a large object that contains many different statistics.

  - **@lengths** The @lengths part is a special way in R to specifically extract (pull out) just the piece of information that lists: “How many receipts had 1 item, how many had 2 items, how many had 3 items,” and so on.

  - **summary.barplot(...)** makes a bar chart showing how many items people bought in each transaction.

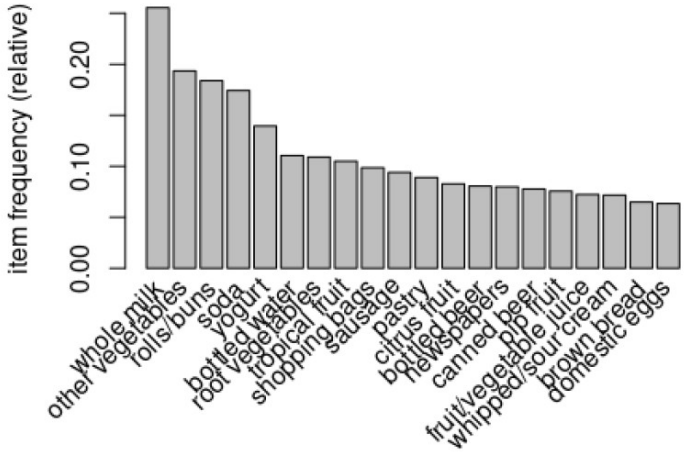
Exploring the Data

```
myGrocIF <- itemFrequency(myGroc)
myGrocIF <- sort(myGrocIF)
```

```
tail(myGrocIF,10)

##          sausage    shopping bags   tropical fruit  root vegetables
##       0.09395018       0.09852567       0.10493137       0.10899847
##    bottled water           yogurt             soda       rolls/buns
##       0.11052364       0.13950178       0.17437722       0.18393493
## other vegetables       whole milk
##       0.19349263       0.25551601
```

```
itemFrequencyPlot(myGroc, topN=20)
```



Explanation:

myGrocIF <- itemFrequency(myGroc)

- This line tells the computer: "Go through every single receipt and count how often each item (like 'milk' or 'bread') appears.

myGrocIF <- sort(myGrocIF)

This line tells the computer:

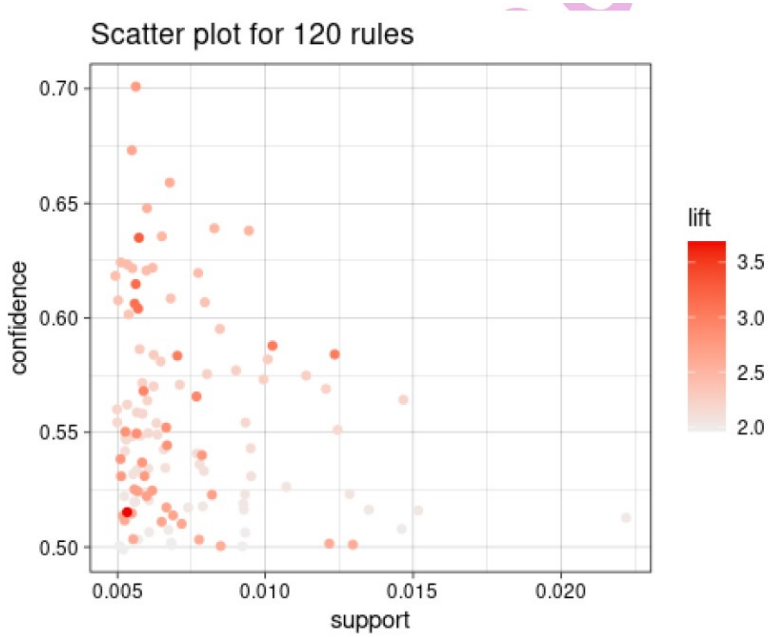
- "Take that list you just made and put it in order from the least popular item at the top, to the most popular item at the bottom."

Model Development

```
#install.packages("arulesViz")

library(arulesViz)

# Now run a test model to examine some rules
ruleset <- apriori(myGroc,
        parameter=list(supp=0.005, conf=0.50),
        control=list(verbose=F))
plot(ruleset)
```



Explanation:

- ❑ **arulesViz** is used for visualizing (drawing) rules.
- ❑ **apriori(myGroc, ...):** Look at your grocery data and find patterns.

Example of a rule: *"If someone buys bread, they also buy milk"*

- ❑ **supp = 0.005 (support)**
  - ○ Only keep rules that happen **at least 0.5% of the time**
  - ○ Very rare patterns are ignored
- ❑ **conf = 0.50 (confidence)**
  - ○ Only keep rules that are **at least 50% correct**
- ❑ **verbose = F**
  - ○ "Don't print too many messages on the screen"

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The End